LABSHEET - 09

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ROLL NO. 03

BTECH CSE - B

```
pip install ultralytics
```

Show hidden output

```
from ultralytics import YOLO
import requests
import cv2
import matplotlib.pyplot as plt

# Image URL
image_url = "https://ultralytics.com/images/bus.jpg"

# Download image
img_data = requests.get(image_url).content
with open("test.jpg", "wb") as f:
    f.write(img_data)

# Load YOLOv8 model
model = YOLO("yolov8n.pt")

# Perform detection
results = model("test.jpg")

# Show result
import cv2
import matplotlib.pyplot as plt

annotated_image = results[0].plot()
annotated_image = cv2.cvtColor(annotated_image, cv2.COLOR_BGR2RGB)

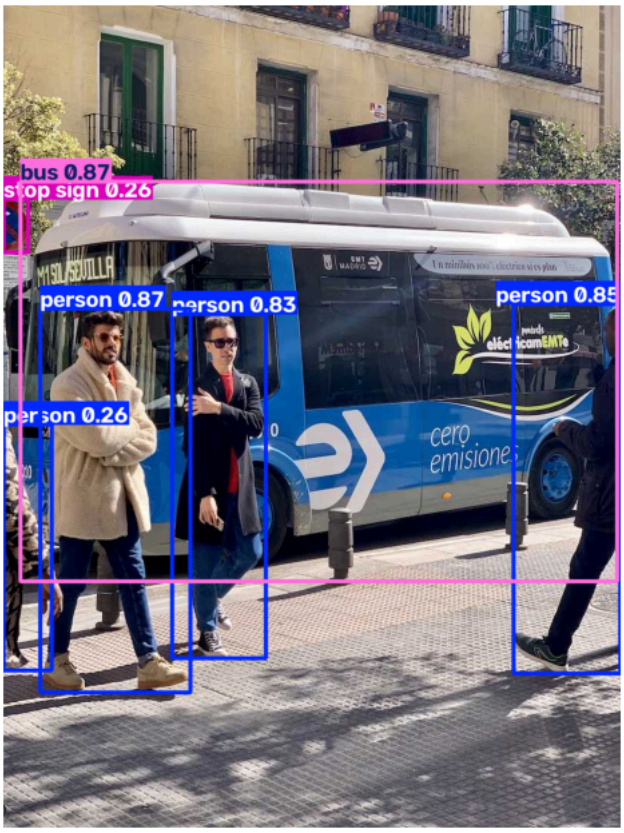
plt.figure(figsize=(10,8))
plt.imshow(annotated_image)
plt.axis("off")
plt.show()

# Print detected objects
for result in results:
    for box in result.boxes:
        cls = int(box.cls[0])
        conf = float(box.conf[0])

        print(f"Object: {model.names[cls]} | Confidence: {conf:.2f}")
```

```
Creating new Ultralytics Settings v0.0.8 file ✅
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see https://docs.ultralytics
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt to 'yolov8n.pt': 100% ━━━━━━━━━━━━ 6.2MB 2

image 1/1 /content/test.jpg: 640x480 4 persons, 1 bus, 1 stop sign, 501.1ms
Speed: 38.0ms preprocess, 501.1ms inference, 40.0ms postprocess per image at shape (1, 3, 640, 480)
```




```
Object: bus | Confidence: 0.87
Object: person | Confidence: 0.87
Object: person | Confidence: 0.85
Object: person | Confidence: 0.83
Object: person | Confidence: 0.26
Object: stop sign | Confidence: 0.26
```

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1. Differentiate between image classification, object detection, and image segmentation.

Ans.Technique Description Output Image Classification Identifies the main object/category in an image. Single class label Object Detection Identifies and locates multiple objects in an image. Class labels + bounding boxes Image Segmentation Assigns a class label to every pixel in the image. Pixel-level object masks

2. Explain the working principle of the YOLO (You Only Look Once) object detection algorithm.

Ans. 2. YOLO treats object detection as a single regression problem. It divides an image into grids and processes the entire image in one pass through a neural network. The network predicts bounding boxes, class probabilities, and confidence scores simultaneously. Non-Maximum Suppression (NMS) is then used to remove duplicate detections.

3. Compare YOLO, SSD, and Faster R-CNN based on speed, accuracy, and practical applications.

Ans. Model Speed Accuracy Applications YOLO Very Fast High Real-time surveillance, autonomous driving SSD Fast Moderate to High Mobile and embedded devices Faster R-CNN Slower Very High Research, medical imaging, high-accuracy tasks

4. What is a bounding box? Why is it important in object detection?

Ans. A bounding box is a rectangular box drawn around an object to indicate its location in an image. It is important because it helps the model localize objects and determine their positions accurately.

5. Explain the significance of confidence score and Intersection over Union (IoU) in object detection.

Ans. Confidence Score: Indicates how certain the model is that an object belongs to a particular class. Intersection over Union (IoU): Measures the overlap between the predicted bounding box and the ground-truth bounding box.

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$

A higher IoU indicates more accurate localization.

6. What are the advantages of using pre-trained deep learning models over traditional image processing techniques?

Ans. Require less training data and time. Provide higher detection accuracy. Can recognize complex object features automatically. Easily adaptable through transfer learning. Robust to variations in lighting, scale, and viewpoint.

7. Why are datasets such as MS COCO and Pascal VOC widely used for object detection research?

Ans. Contain large numbers of labeled images. Include multiple object categories. Provide standardized benchmarks for evaluation. Support fair comparison between different models. Represent diverse real-world scenarios.

8. Mention five real-world applications of object detection in computer vision. Autonomous vehicles.

Ans.Video surveillance and security monitoring. Medical image analysis. Retail analytics and inventory management. Industrial quality inspection.

9. What challenges are commonly encountered while detecting objects in complex real-world environments?

Ans. Occlusion (objects partially hidden). Variations in lighting and weather conditions. Small object detection. Complex backgrounds. Object scale and orientation changes. Real-time processing requirements.

10. How can object detection systems be further improved using recent advancements in deep learning and computer vision?

Ans. Using transformer-based architectures (e.g., DETR). Training on larger and more diverse datasets. Applying data augmentation techniques. Using ensemble models for better accuracy. Leveraging edge AI and hardware acceleration. Incorporating self-supervised and transfer learning methods.